# P8106_Final

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```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   4.0.0     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.1.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(caret)
```

```
## Loading required package: lattice
##
## Attaching package: 'caret'
##
## The following object is masked from 'package:purrr':
##
##     lift
```

```
library(tidymodels)
```

```
## -- Attaching packages --------------------------------------- tidymodels 1.4.1 --
## v broom        1.0.9     v rsample      1.3.2
## v dials        1.4.2     v tailor       0.1.0
## v infer        1.0.9     v tune         2.0.1
## v modeldata    1.5.1     v workflows    1.3.0
## v parsnip      1.4.1     v workflowsets 1.1.1
## v recipes      1.3.1     v yardstick    1.3.2

## Warning: package 'parsnip' was built under R version 4.5.2

## Warning: package 'rsample' was built under R version 4.5.2

## -- Conflicts ------------------------------------------ tidymodels_conflicts() --
## x rsample::calibration()   masks caret::calibration()
```

```
## x scales::discard()        masks purrr::discard()
## x dplyr::filter()          masks stats::filter()
## x recipes::fixed()         masks stringr::fixed()
## x dplyr::lag()             masks stats::lag()
## x caret::lift()            masks purrr::lift()
## x yardstick::precision()   masks caret::precision()
## x yardstick::recall()      masks caret::recall()
## x yardstick::sensitivity() masks caret::sensitivity()
## x yardstick::spec()        masks readr::spec()
## x yardstick::specificity() masks caret::specificity()
## x recipes::step()          masks stats::step()
```

```
library(gbm)
```

```
## Warning: package 'gbm' was built under R version 4.5.2
```

```
## Loaded gbm 2.2.3
## This version of gbm is no longer under development. Consider transitioning to gbm3, https://github.c
```

```
library(e1071)
```

```
##
## Attaching package: 'e1071'
##
## The following object is masked from 'package:tune':
##
##     tune
##
## The following object is masked from 'package:rsample':
##
##     permutations
##
## The following object is masked from 'package:parsnip':
##
##     tune
##
## The following object is masked from 'package:ggplot2':
##
##     element
```

```
library(ranger)
library(pROC)
```

```
## Type 'citation("pROC")' for a citation.
##
## Attaching package: 'pROC'
##
## The following objects are masked from 'package:stats':
##
##     cov, smooth, var
```

```r
library(pdp)
```

```
## Warning: package 'pdp' was built under R version 4.5.2
```

```
##
## Attaching package: 'pdp'
##
## The following object is masked from 'package:purrr':
##
##     partial
```

```r
library(patchwork)
library(gridExtra)
```

```
##
## Attaching package: 'gridExtra'
##
## The following object is masked from 'package:dplyr':
##
##     combine
```

```r
library(corrplot)
```

```
## corrplot 0.95 loaded
```

```r
library(ggplot2)
```

## Load the data

```r
load("severity.RData")
dat <- na.omit(dat)
set.seed(4027)
my_dat <- dat[sample(nrow(dat), 1000), ]
```

## Convert Variables

```r
my_dat <- my_dat %>%
  mutate(
    severity = factor(severity,
                      levels = c(0,1),
                      labels = c("NotSevere","Severe")),

    gender = factor(gender,
                    levels = c(0,1),
                    labels = c("Female","Male")),
```

```
    hypertension = factor(hypertension,
                          levels = c(0,1),
                          labels = c("No","Yes")),
    diabetes = factor(diabetes,
                      levels = c(0,1),
                      labels = c("No","Yes")),
    vaccine = factor(vaccine,
                     levels = c(0,1),
                     labels = c("Not vaccinated","Vaccinated")),

    race = factor(race,
                  levels = c(1,2,3,4),
                  labels = c("White","Asian","Black","Hispanic")),

    smoking = factor(smoking,
                     levels = c(0,1,2),
                     labels = c("Never","Former","Current"))
  )
```

## Train test split

```
set.seed(1234)
split <- initial_split(my_dat,
                       prop = 0.8,
                       strata = severity)

train_dat <- training(split)
test_dat  <- testing(split)
```

## Exploratory Analysis

```
# Summary table
summary_table <- train_dat %>%
  select(-id) %>%
  summary()
summary_table
```

```
##       age           gender          race        smoking        height
##  Min.   :48.00   Female:408   White   :491   Never  :469   Min.   :149.2
##  1st Qu.:57.00   Male  :391   Asian   : 46   Former :242   1st Qu.:165.8
##  Median :60.00                Black   :172   Current: 88   Median :170.0
##  Mean   :60.17                Hispanic: 90                 Mean   :169.7
##  3rd Qu.:63.00                                             3rd Qu.:173.9
##  Max.   :74.00                                             Max.   :186.5
##      weight           bmi        diabetes  hypertension      SBP
##  Min.   :58.40   Min.   :20.5   No :688   No :408      Min.   :101.0
##  1st Qu.:75.35   1st Qu.:26.0   Yes:111   Yes:391      1st Qu.:124.0
```

```
##  Median :80.20   Median :27.6                          Median :130.0
##  Mean   :79.93   Mean   :27.8                          Mean   :130.1
##  3rd Qu.:84.70   3rd Qu.:29.5                          3rd Qu.:136.0
##  Max.   :98.10   Max.   :36.8                          Max.   :159.0
##       LDL                 vaccine      depression         severity
##  Min.   : 52.0   Not vaccinated:303   Min.   : 0.000   NotSevere:546
##  1st Qu.: 96.0   Vaccinated    :496   1st Qu.: 6.000   Severe   :253
##  Median :111.0                        Median : 7.000
##  Mean   :109.9                        Mean   : 6.996
##  3rd Qu.:123.0                        3rd Qu.: 8.000
##  Max.   :178.0                        Max.   :13.000
```

```r
# Continuous variable
p1 <- ggplot(my_dat,
             aes(x = severity,
                 y = age,
                 fill = severity)) +
  geom_boxplot() +
  theme_bw() +
  ggtitle("Age by Severity")

p2 <- ggplot(my_dat,
             aes(x = severity,
                 y = height,
                 fill = severity)) +
  geom_boxplot() +
  theme_bw() +
  ggtitle("Height by Severity")

p3 <- ggplot(my_dat,
             aes(x = severity,
                 y = weight,
                 fill = severity)) +
  geom_boxplot() +
  theme_bw() +
  ggtitle("Weight by Severity")

p4 <- ggplot(my_dat,
             aes(x = severity,
                 y = bmi,
                 fill = severity)) +
  geom_boxplot() +
  theme_bw() +
  ggtitle("BMI by Severity")

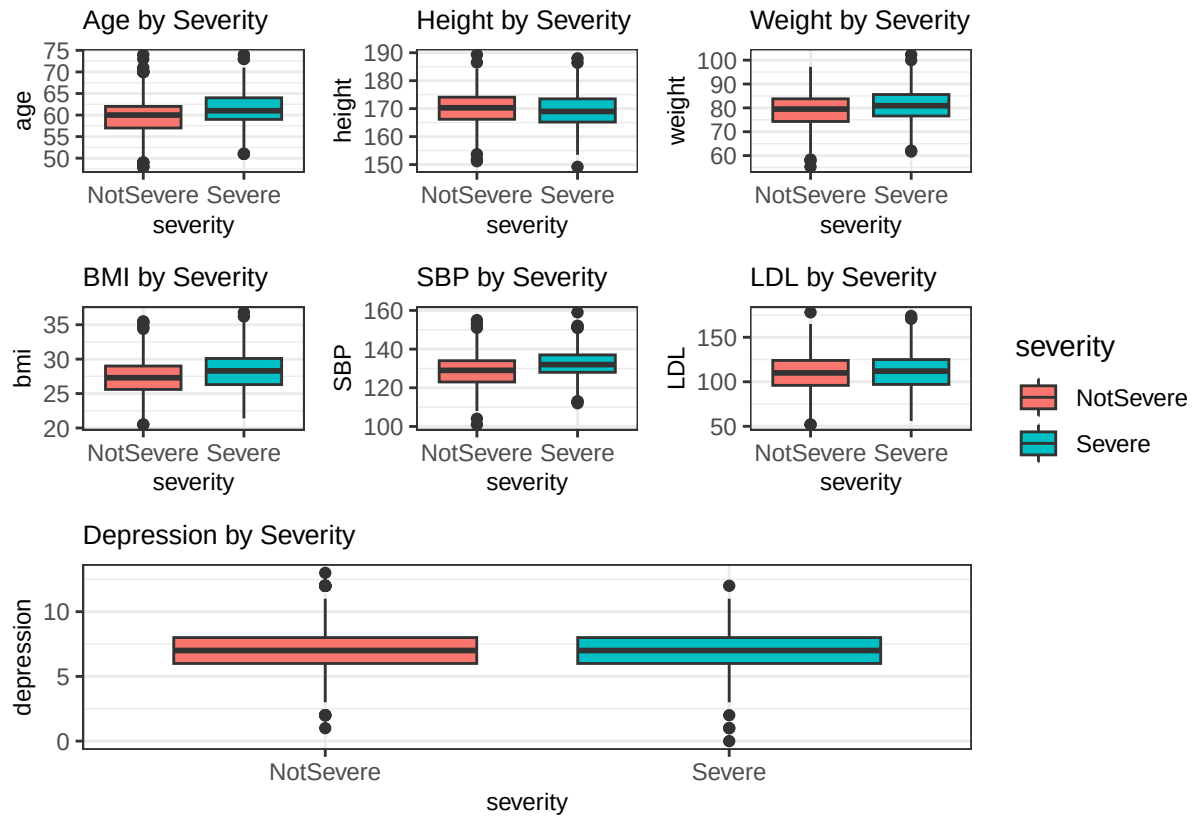
p5 <- ggplot(my_dat,
             aes(x = severity,
                 y = SBP,
                 fill = severity)) +
  geom_boxplot() +
  theme_bw() +
  ggtitle("SBP by Severity")

p6 <- ggplot(my_dat,
```

```
              aes(x = severity,
                  y = LDL,
                  fill = severity)) +
  geom_boxplot() +
  theme_bw() +
  ggtitle("LDL by Severity")

p7 <- ggplot(my_dat,
              aes(x = severity,
                  y = depression,
                  fill = severity)) +
  geom_boxplot() +
  theme_bw() +
  ggtitle("Depression by Severity")

((p1 + p2 + p3) /
 (p4 + p5 + p6) /
 (p7)) +
plot_layout(guides = "collect",
            heights = c(1,1,1.5)) &
theme(
  plot.title = element_text(size = 10),
  axis.title = element_text(size = 9),
  axis.text = element_text(size = 9)
)
```



```r
# Categorical variables
c1 <- ggplot(my_dat,
             aes(x = gender,
                 fill = severity)) +
  geom_bar(position = "fill") +
  theme_bw() +
  ggtitle("Gender vs Severity")

c2 <- ggplot(my_dat,
             aes(x = race,
                 fill = severity)) +
  geom_bar(position = "fill") +
  theme_bw() +
  ggtitle("Race vs Severity")

c3 <- ggplot(my_dat,
             aes(x = smoking,
                 fill = severity)) +
  geom_bar(position = "fill") +
  theme_bw() +
  ggtitle("Smoking vs Severity")

c4 <- ggplot(my_dat,
             aes(x = hypertension,
                 fill = severity)) +
  geom_bar(position = "fill") +
  theme_bw() +
  ggtitle("Hypertension vs Severity")

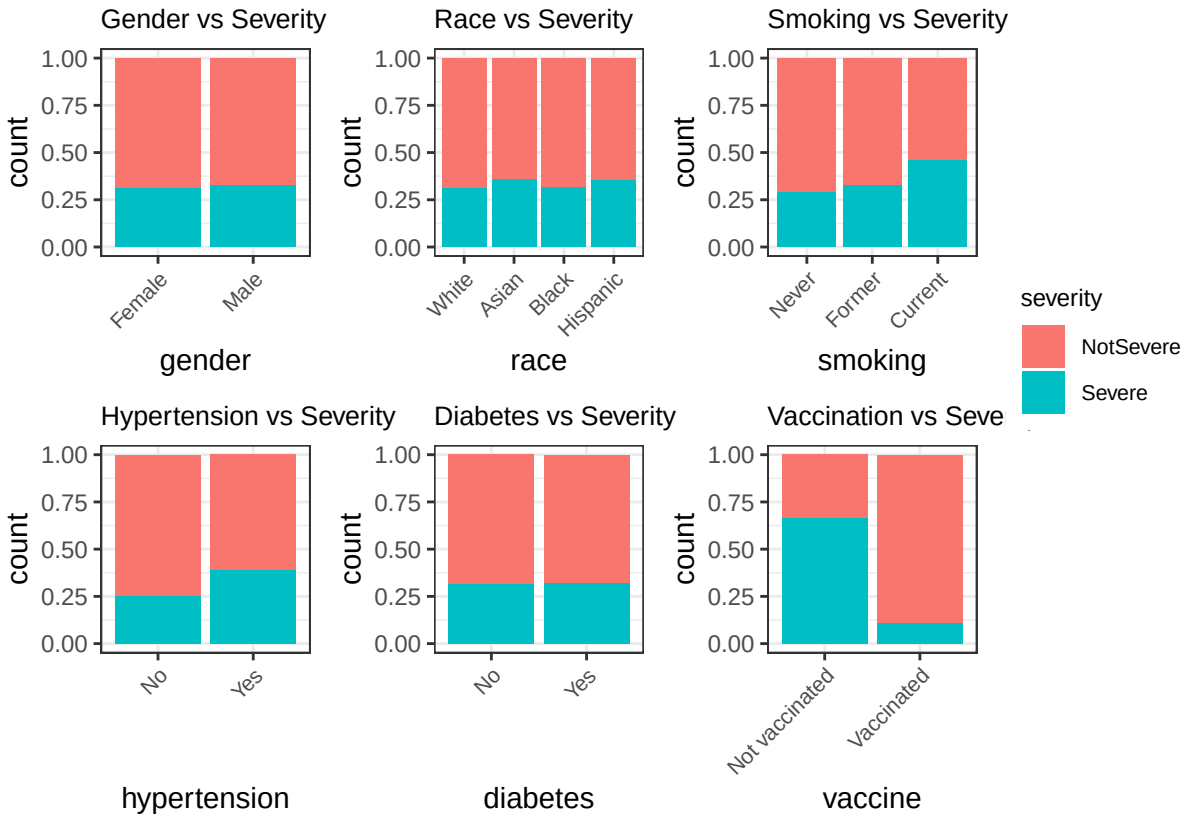

c5 <- ggplot(my_dat,
             aes(x = diabetes,
                 fill = severity)) +
  geom_bar(position = "fill") +
  theme_bw() +
  ggtitle("Diabetes vs Severity")


c6 <- ggplot(my_dat,
             aes(x = vaccine,
                 fill = severity)) +
  geom_bar(position = "fill") +
  theme_bw() +
  ggtitle("Vaccination vs Severity")

common_theme <- theme_bw() +
  theme(
    axis.text.x = element_text(angle = 45,
                               hjust = 1,
                               size = 8),
    plot.title = element_text(size = 10),
    legend.title = element_text(size = 9),
    legend.text = element_text(size = 8)
```

```
  )

((c1 + c2 + c3) /
(c4 + c5 + c6)) +
  plot_layout(guides = "collect") &
  common_theme
```

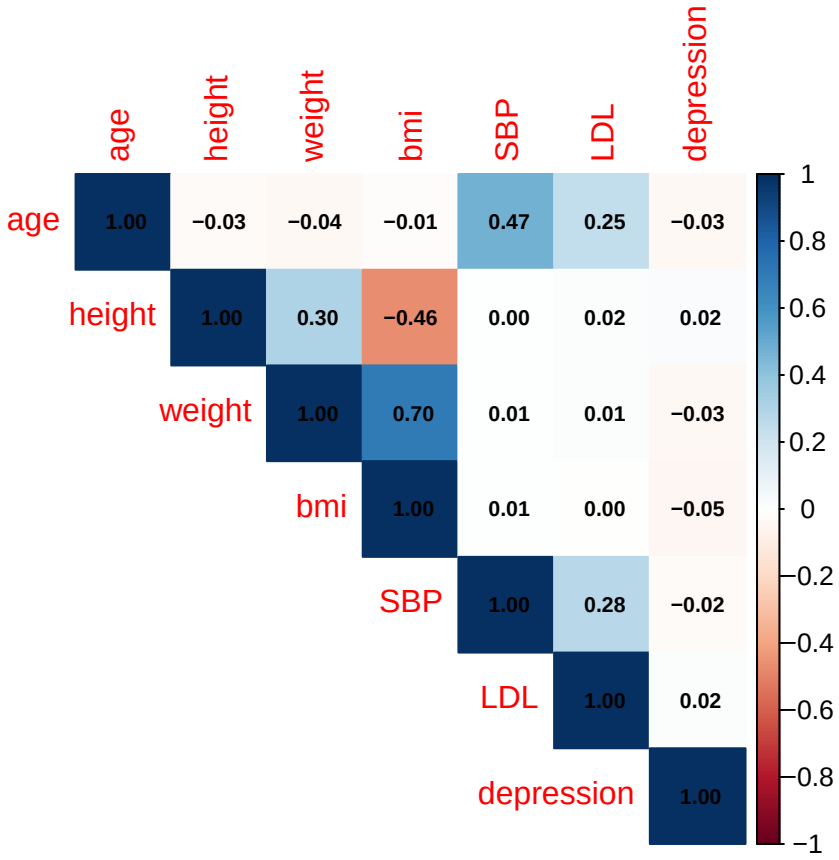



```
# Correlations
continuous_vars <- my_dat %>%
  select(
    age,
    height,
    weight,
    bmi,
    SBP,
    LDL,
    depression
  )

cor_matrix <- cor(continuous_vars)

corrplot(
  cor_matrix,
  method = "color",
  type = "upper",
```

```
  addCoef.col = "black",
  number.cex = 0.7
)
```




## Logistic Regression

```
logit_fit <- glm(
  severity ~ age + gender + race + smoking +
    bmi + hypertension + diabetes +
    SBP + LDL + vaccine + depression,
  data = train_dat,
  family = binomial
)

summary(logit_fit)
```

```
##
## Call:
## glm(formula = severity ~ age + gender + race + smoking + bmi +
##     hypertension + diabetes + SBP + LDL + vaccine + depression,
##     family = binomial, data = train_dat)
##
```

```
## Coefficients:
##                      Estimate Std. Error z value Pr(>|z|)
## (Intercept)       -23.391176   3.365722  -6.950 3.66e-12 ***
## age                 0.090661   0.027512   3.295 0.000983 ***
## genderMale         -0.015373   0.210266  -0.073 0.941715
## raceAsian           0.273030   0.470198   0.581 0.561463
## raceBlack          -0.065149   0.258748  -0.252 0.801206
## raceHispanic       -0.102350   0.343499  -0.298 0.765731
## smokingFormer       0.330010   0.233271   1.415 0.157154
## smokingCurrent      1.114327   0.333848   3.338 0.000844 ***
## bmi                 0.260503   0.043767   5.952 2.65e-09 ***
## hypertensionYes    -0.351095   0.350140  -1.003 0.315992
## diabetesYes         0.141854   0.288059   0.492 0.622402
## SBP                 0.099607   0.024021   4.147 3.37e-05 ***
## LDL                -0.008684   0.005437  -1.597 0.110243
## vaccineVaccinated  -3.527781   0.249279 -14.152  < 2e-16 ***
## depression         -0.071642   0.051379  -1.394 0.163200
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 997.66  on 798  degrees of freedom
## Residual deviance: 598.93  on 784  degrees of freedom
## AIC: 628.93
##
## Number of Fisher Scoring iterations: 5
```

```r
# OR
exp(cbind(
  OR = coef(logit_fit),
  confint(logit_fit)
))
```

```
## Waiting for profiling to be done...

##                             OR        2.5 %       97.5 %
## (Intercept)       6.939707e-11 7.812562e-14 4.291448e-08
## age               1.094898e+00 1.037957e+00 1.156346e+00
## genderMale        9.847442e-01 6.510128e-01 1.486461e+00
## raceAsian         1.313940e+00 5.157043e-01 3.273127e+00
## raceBlack         9.369275e-01 5.619677e-01 1.552268e+00
## raceHispanic      9.027136e-01 4.560450e-01 1.757536e+00
## smokingFormer     1.390982e+00 8.804538e-01 2.200539e+00
## smokingCurrent    3.047516e+00 1.588231e+00 5.893067e+00
## bmi               1.297583e+00 1.193052e+00 1.416724e+00
## hypertensionYes   7.039169e-01 3.528559e-01 1.395435e+00
## diabetesYes       1.152409e+00 6.521443e-01 2.021667e+00
## SBP               1.104736e+00 1.054744e+00 1.159061e+00
## LDL               9.913535e-01 9.808131e-01 1.001979e+00
## vaccineVaccinated 2.937001e-02 1.767488e-02 4.705515e-02
## depression        9.308639e-01 8.410008e-01 1.029013e+00
```

```r
# Predicted Probabilities
logit_prob <- predict(
  logit_fit,
  newdata = test_dat,
  type = "response"
)
```

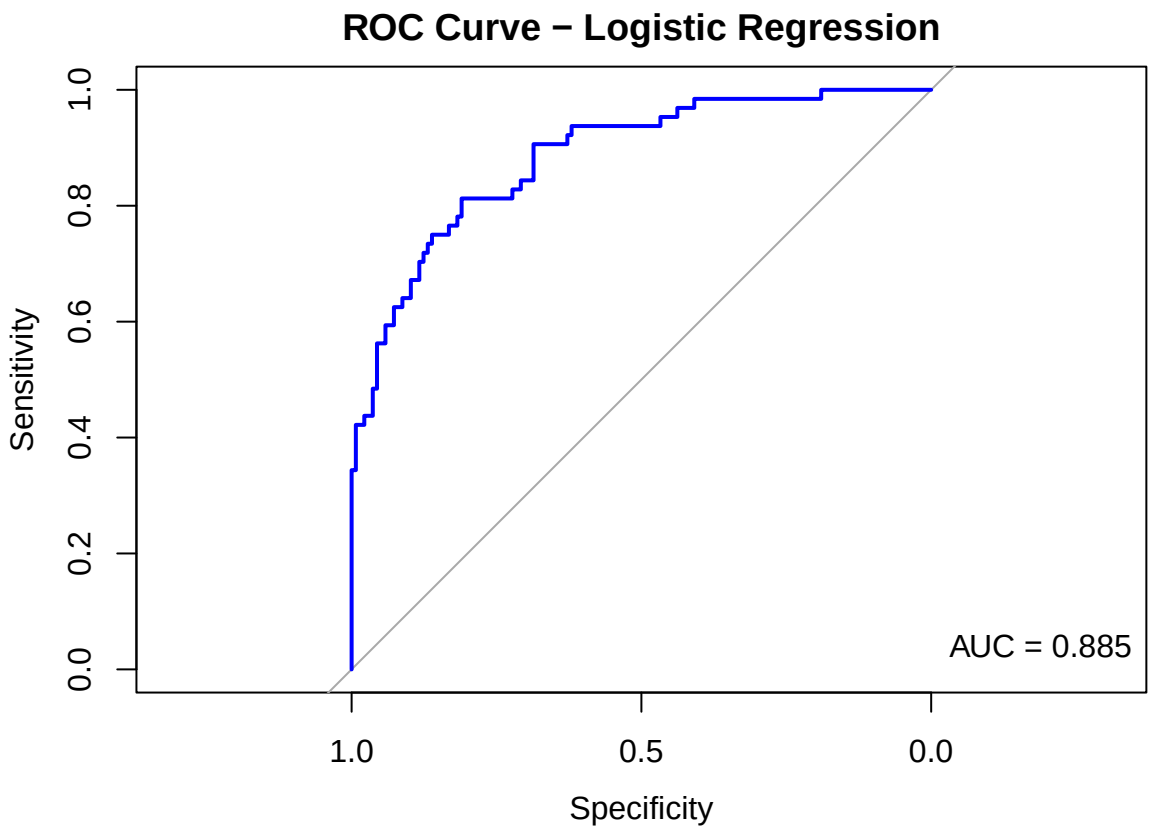
```r
# ROC
roc_logit <- roc(
  response = test_dat$severity,
  predictor = logit_prob,
  levels = c("NotSevere","Severe")
)
```

```
## Setting direction: controls < cases
```

```r
auc_logit <- auc(roc_logit)

plot(
  roc_logit,
  main = "ROC Curve - Logistic Regression",
  col = "blue",
  lwd = 2
)

legend(
  "bottomright",
  legend = paste("AUC =", round(auc_logit, 3)),
  bty = "n"
)
```



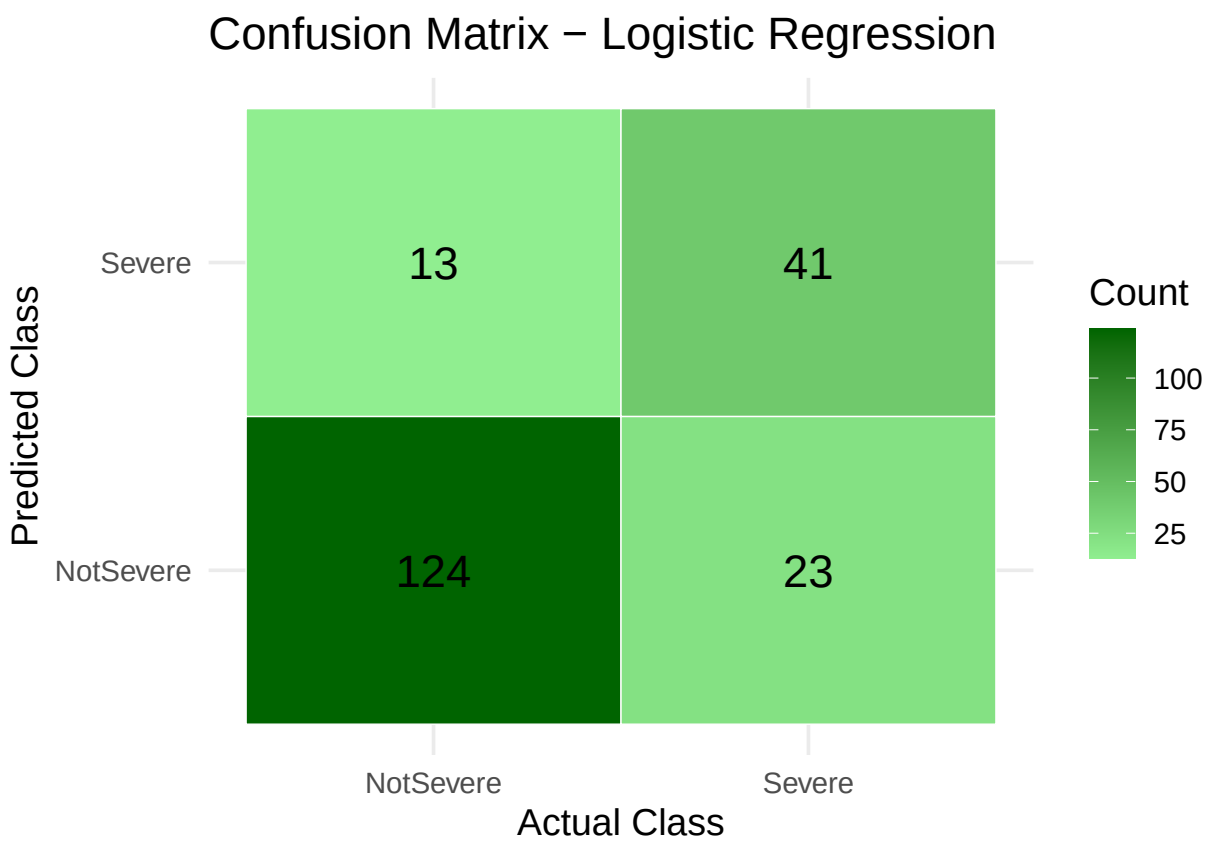

```
# Predictions
logit_pred <- ifelse(logit_prob > 0.5,
                     "Severe",
                     "NotSevere")

logit_pred <- factor(
  logit_pred,
  levels = c("NotSevere","Severe")
)

# Confusion matrix
cm <- confusionMatrix(
  data = logit_pred,
  reference = test_dat$severity,
  positive = "Severe"
)

cm_df <- as.data.frame(cm$table)
# Plot
ggplot(cm_df, aes(x = Reference, y = Prediction, fill = Freq)) +
  geom_tile(color = "white") +
  geom_text(aes(label = Freq), size = 6) +
  scale_fill_gradient(low = "lightgreen", high = "darkgreen") +
  labs(
    title = "Confusion Matrix - Logistic Regression",
    x = "Actual Class",
```

```
    y = "Predicted Class",
    fill = "Count"
  ) +
  theme_minimal(base_size = 14)
```




## Boosting

```
ctrl <- trainControl(method = "cv", number = 5)
```

```
# tuning grid
gbm.grid <- expand.grid(
  n.trees = c(100, 200, 500, 1000, 2000, 5000),
  interaction.depth = 1:4,
  shrinkage = c(0.005, 0.01, 0.05),
  n.minobsinnode = 10
)
```

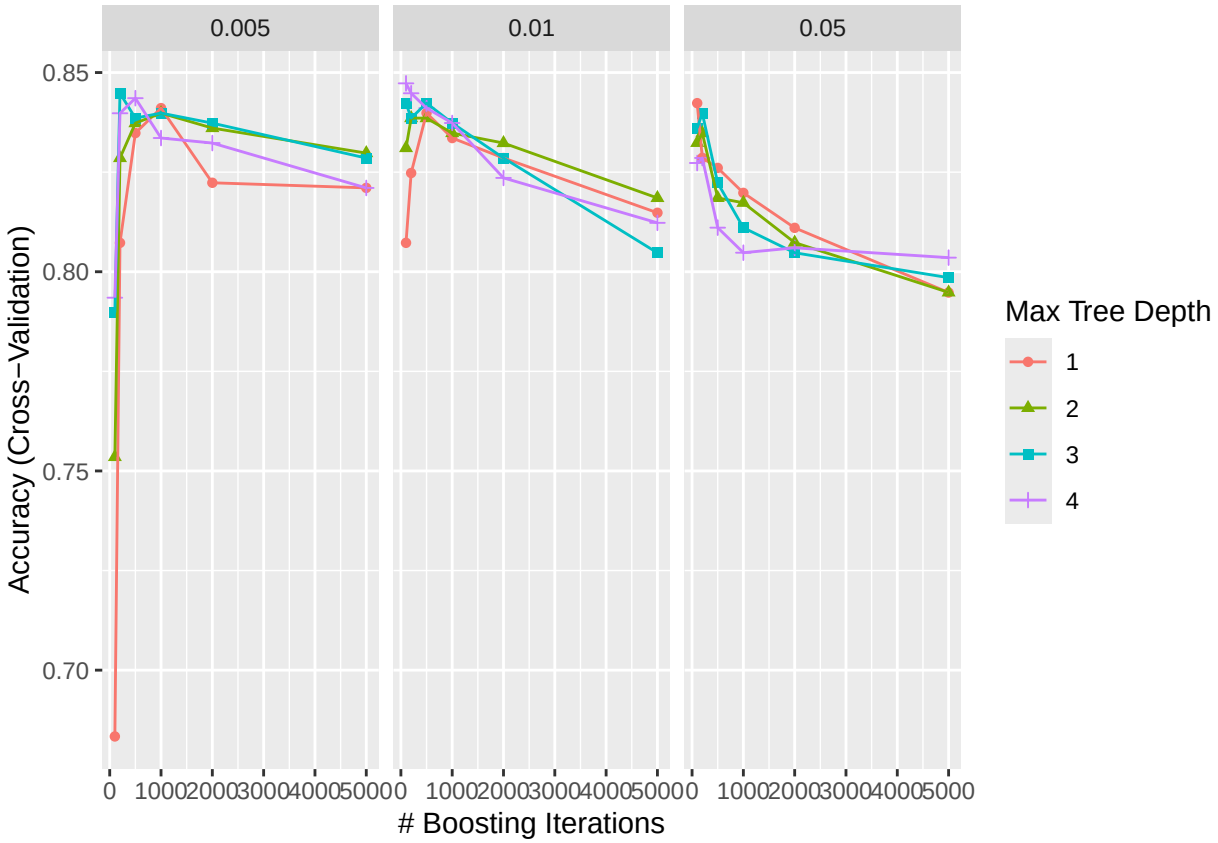
```
# Train
set.seed(4027)

gbm_fit <- train(
  severity ~ . -id,
```

```
  data = train_dat,
  method = "gbm",
  tuneGrid = gbm.grid,
  trControl = ctrl,
  verbose = FALSE
)
```

```
ggplot(gbm_fit)
```

```
## Warning: `aes_string()` was deprecated in ggplot2 3.0.0.
## i Please use tidy evaluation idioms with `aes()`.
## i See also `vignette("ggplot2-in-packages")` for more information.
## i The deprecated feature was likely used in the caret package.
##   Please report the issue at <https://github.com/topepo/caret/issues>.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```




```
# Best parameters
gbm_fit$bestTune
```

```
##    n.trees interaction.depth shrinkage n.minobsinnode
## 43     100                 4      0.01             10
```

```r
gbm_prob <- predict(
  gbm_fit,
  newdata = test_dat,
  type = "prob"
)[,"Severe"]
```

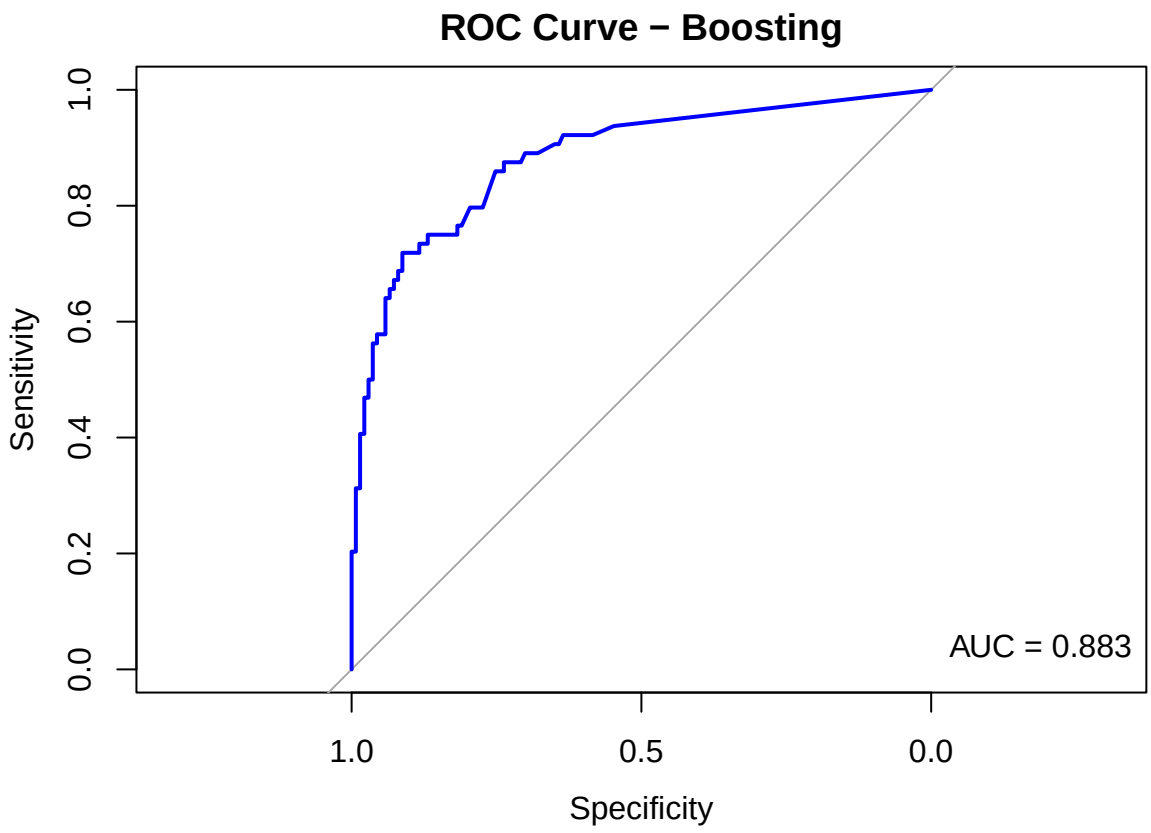
```r
roc_gbm <- roc(
  response = test_dat$severity,
  predictor = gbm_prob,
  levels = c("NotSevere","Severe")
)
```

```
## Setting direction: controls < cases
```

```r
auc_gbm <- auc(roc_gbm)

plot(
  roc_gbm,
  main = "ROC Curve - Boosting",
  col = "blue",
  lwd = 2
)

legend(
  "bottomright",
  legend = paste(
    "AUC =",
    round(auc_gbm, 3)
  ),
  bty = "n"
)
```



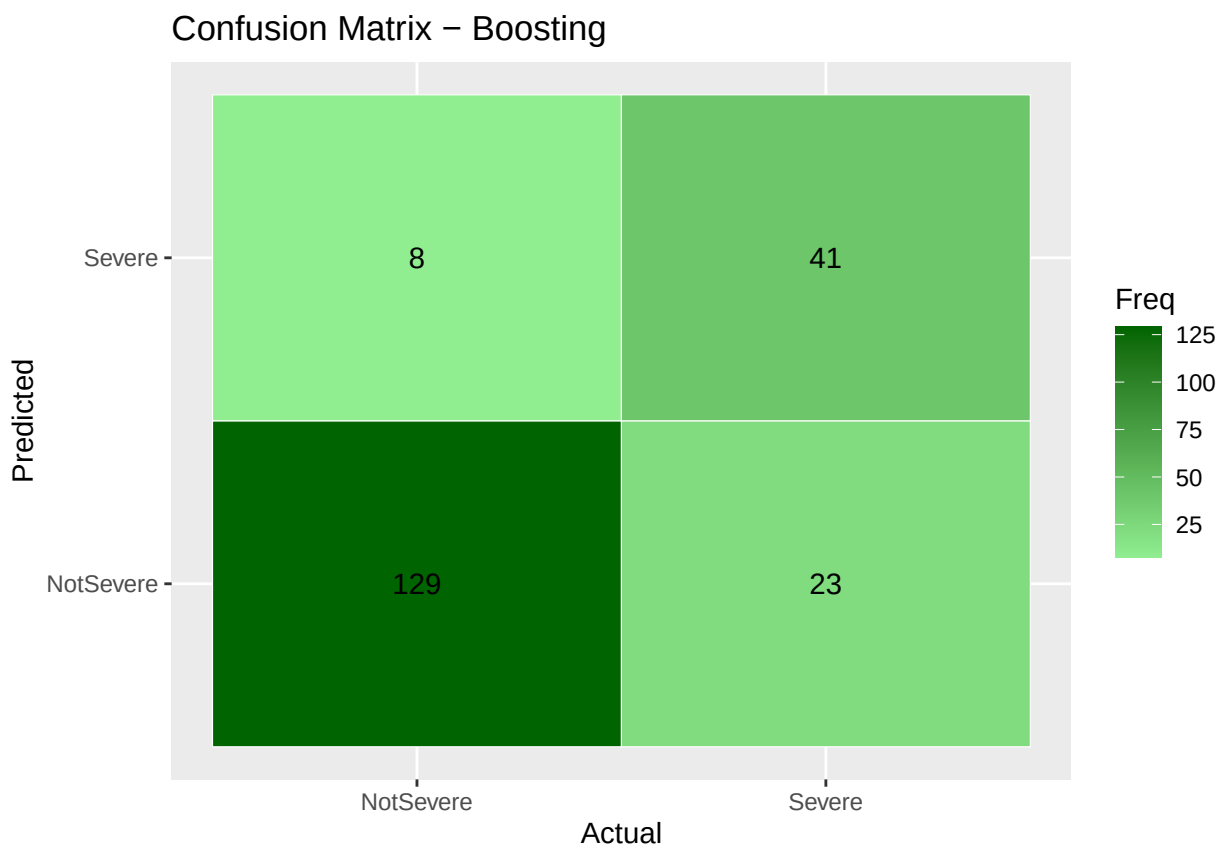

```r
gbm_pred <- predict(
  gbm_fit,
  newdata = test_dat
)

cm <- confusionMatrix(
  data = gbm_pred,
  reference = test_dat$severity
)

cm_df <- as.data.frame(cm$table)

ggplot(
  cm_df,
  aes(
    x = Reference,
    y = Prediction,
    fill = Freq
  )
) +
  geom_tile(color = "white") +
  geom_text(
    aes(label = Freq)
  ) +
  scale_fill_gradient(
    low = "lightgreen",
```

```
    high = "darkgreen"
  ) +
  labs(
    title = "Confusion Matrix - Boosting",
    x = "Actual",
    y = "Predicted"
  )
```

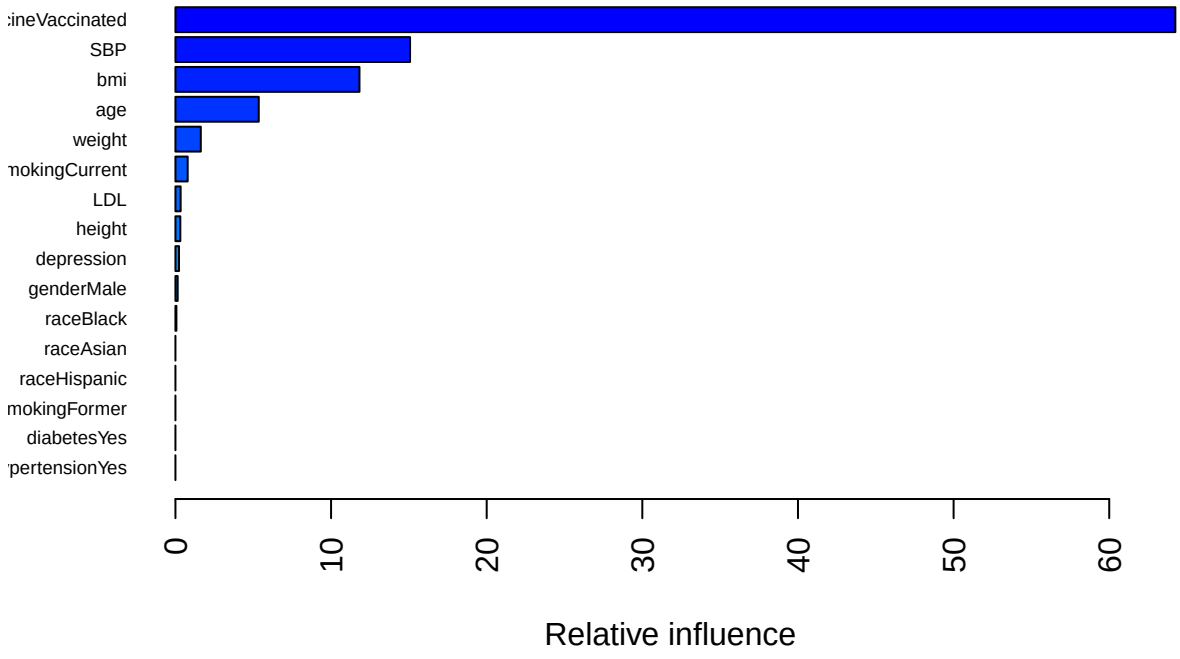



```
varImp(gbm_fit)
```

```
## gbm variable importance
##
##                     Overall
## vaccineVaccinated 100.00000
## SBP                23.46231
## bmi                18.40310
## age                 8.33070
## weight              2.53702
## smokingCurrent      1.22403
## LDL                 0.52045
## height              0.48868
## depression          0.35782
## genderMale          0.22290
## raceBlack           0.09565
## hypertensionYes     0.00000
```

```
## raceAsian           0.00000
## diabetesYes         0.00000
## raceHispanic        0.00000
## smokingFormer       0.00000
```

```
summary(gbm_fit$finalModel, las = 2, cBars = 19, cex.names = 0.6)
```




```
##                                 var     rel.inf
## vaccineVaccinated vaccineVaccinated 64.24973958
## SBP                             SBP 15.07447302
## bmi                             bmi 11.82394513
## age                             age  5.35245077
## weight                       weight  1.63003164
## smokingCurrent       smokingCurrent  0.78643380
## LDL                             LDL  0.33438564
## height                       height  0.31397817
## depression               depression  0.22989537
## genderMale               genderMale  0.14321037
## raceBlack                 raceBlack  0.06145652
## raceAsian                 raceAsian  0.00000000
## raceHispanic           raceHispanic  0.00000000
## smokingFormer         smokingFormer  0.00000000
## diabetesYes             diabetesYes  0.00000000
## hypertensionYes     hypertensionYes  0.00000000
```